Environmental Improvement Assessment: Assessment and Prediction of Changes in Urban Environmental Quality in China after The Implementation of The New Regulations on Plastic Restrictions

Abstract: This article aims to evaluate the changes and trends in urban environmental quality after the implementation of the new regulations of Chinese plastic restriction order in 2020, and evaluate its effectiveness by analyzing the data changes in urban environmental quality presented by the implementation of the plastic restriction order. Through the comparison and analysis of China's urban environmental quality data before and after the change, we can determine whether the implementation of the new regulations of plastic restriction has really promoted the improvement of environmental quality. This study collected data on urban environmental quality indicators before and after the implementation of the new plastic restriction regulations, and compared and analyzed the trend of environmental quality changes. The results show that within a short period of time after the implementation of the new plastic restriction regulations, there have been significant positive changes in urban environmental indicator data in China, especially in cities with significant restrictions. After the implementation of the new regulations on plastic restriction, the overall quality of urban environment in China has been improved, including indicators such as air, water, and noise. Especially in large and medium-sized cities, the improvement in environmental quality is more significant, while in small cities, the improvement is relatively small. According to the model prediction, this trend of environmental quality improvement will continue in the coming years. This article believes that the implementation of the new plastic restriction regulations is one of the important factors in improving the environmental quality of Chinese cities, and it has played a huge role in environmental governance.

Keywords: urban environmental quality; Chinese plastic restriction; environmental governance; indicators

1. Introduction

1.1. Background and importance of the problem

Plastic pollution has become one of the global problems, and the massive accumulation of plastic waste threatens the human ecological environment and health. The accumulation of plastic waste in environments such as land, rivers, and oceans has seriously damaged water resources, soil fertility, and biodiversity. At the same time, the long-term decomposition process of plastic waste will also exacerbate the persistent pollution to the global environment. Therefore, the treatment of Plastic pollution has very important theoretical and practical significance. For China, where urbanization is deepening and plastic waste generation and Plastic pollution are increasingly prominent, the implementation of the plastic restriction order is closely related to urban environmental quality. It is related to people's life and health, and has certain theoretical and practical value for improving urban environmental protection awareness and improving urban environmental quality.

1.2. Research purpose and significance

With the acceleration of urbanization, Chinese cities are facing increasingly severe environmental pollution problems. The problem of Plastic pollution not only has a negative impact on the life of urban residents, but also has damaged the health of the ecosystem. The implementation goal of the plastic restriction order is to protect the environmental ecosystem, limit the production, use, and treatment of plastic products, and reduce the harm of plastic waste to the environment. The implementation goal of the plastic restriction order is also to improve the quality of the urban environment. By limiting the generation and treatment of plastic waste, urban environmental quality has been improved, and key environmental indicators such as $PM_{2.5}$, PM_{10}, O_3, SO_2, NO_2 and CO have been significantly improved. This helps to improve the quality of life of urban residents and maintain the sustainable development of the city's economy and society.

With a deeper understanding of environmental issues, governments around the world have begun to take proactive environmental measures to minimize environmental pressure and protect public health. Among them, plastic restriction orders, as a proactive and normalized measure, not only made important contributions to environmental protection, but also led to the emergence of similar policies. China officially launched its first national plastic restriction order in 2008, and subsequent central documents have clarified the government's specific guidance on the treatment of waste plastics and the recycling and treatment of plastic waste. This study is based on China's 2020 "Opinions of the National Development and Reform Commission and the Ministry of Ecological Environment on Further Strengthening the Treatment of Plastic pollution", focusing on its impact on urban environmental quality, and discussing its future trends. The purpose of this article is to comprehensively and objectively evaluate the changes in urban environmental quality in China after the implementation of the plastic restriction order, and to study its cost-effectiveness, future impact, and development based on this.

Figure 1. Opinions of the National Development and Reform Commission and the Ministry of Ecology and Environment on Further Strengthening the Control of Plastic Pollution. (2020).

国家发展改革委 生态环境部关于
进一步加强塑料污染治理的意见

发改环资〔2020〕80号

各省、自治区、直辖市人民政府，国务院各部委、各直属机构：

塑料在生产生活中应用广泛，是重要的基础材料。不规范生产、使用塑料制品和回收处置塑料废弃物，会造成能源资源浪费和环境污染，加大资源环境压力。积极应对塑料污染，事关人民群众健康，事关我国生态文明建设和高质量发展。为贯彻落实党中央、国务院决策部署，进一步加强塑料污染治理，建立健全塑料制品长效管理机制，经国务院同意，现提出如下意见。

一、总体要求

（一）指导思想。以习近平新时代中国特色社会主义思想为指导，全面贯彻党的十九大和十九届二中、三中、四中全会精神，坚持以人民为中心，牢固树立新发展理念，有序禁止、限制部分塑料制品的生产、销售和使用，积极推广替代产品，规范塑料废弃物回收利用，建立健全塑料制品生产、流通、使用、回收处置等环节的管理制度，有力有序有效治理塑料污染，努力建设美丽中国。

（二）基本原则。

突出重点，有序推进。强化源头治理，抓住塑料制品生产使用的重点领域和重要环节，针对社会反映强烈的突出问题，分类提出管理要求；综合考虑各地区、各领域实际情况，合理确定实施路径，积极稳妥推进塑料污染治理工作。

创新引领，科技支撑。以可循环、易回收、可降解为导向，研发推广性能达标、绿色环保、经济适用的塑料制品及替代产品，培育有利于规范回收和循环利用、减少塑料污染的新业态新模式。

多元参与，社会共治。发挥企业主体责任，强化政府监督管理，加强政策引导，凝聚社会共识，形成政府、企业、行业组织、社会公众共同参与的多元共治体系。

（三）主要目标。到2020年，率先在部分地区、部分领域禁止、限制部分塑料制品的生产、销售和使用。到2022年，一次性塑料制品消费量明显减少，替代产品得到推广，塑料废弃物资源化能源化利用比例大幅提升；在塑料污染问题突出领域和电商、快递、外卖等新兴领域，形成一批可复制、可推广的塑料减量和绿色物流模式。到2025年，塑料制品生产、流通、消费和回收处置等环节的管理制度基本建立，多元共治体系基本形成，替代产品开发应用水平进一步提升，重点城市塑料垃圾填埋量大幅降低，塑料污染得到有效控制。

The text will conduct a comprehensive and systematic assessment of the changes in urban environmental quality in China after the implementation of the new plastic restriction order, in order to determine its environmental benefits and cost-effectiveness, and provide data support for future formulation of urban environmental protection policies. Specifically, we investigated whether the implementation of the new regulations on plastic restriction has truly promoted the improvement of urban environmental quality, evaluated its environmental benefits and cost-effectiveness, and explored its future impact on the development and protection of China's urban environment.

This article will explore the actual situation of the urban environment in China after the implementation of the new plastic restriction regulations? Has the implementation of plastic restrictions truly promoted the improvement of urban environmental quality? Is this effect worth its cost and long-term execution? What is the trend of changes in urban environmental quality before and after the implementation of the plastic restriction order? By studying and solving these problems, we can analyze the trend and effectiveness of data changes in urban environmental quality in China after the implementation of the plastic restriction order, and explore its potential value in environmental protection.

2. Research method

2.1. Related data analysis

This study adopts a horizontal comparative research design to collect, sort out and analyze the urban environmental quality data of China at two time points (around 2020) before and after the implementation of the plastic restriction order. Firstly, we obtained data indicators on urban air quality, water quality, and waste treatment from 2016 to 2022 through official channels such as the website of the Bureau of Statistics and the website of the Environmental Protection Department, including $PM_{2.5}$, PM_{10}, O_3, SO_2, NO_2, CO, etc. These indicators are key factors for evaluating urban environmental quality, with objective and scientific nature, and can reflect the impact of plastic restriction on urban environmental quality.

Figure 2. Data from the Ministry of Ecology and Environment.

长三角地区六项污染物浓度

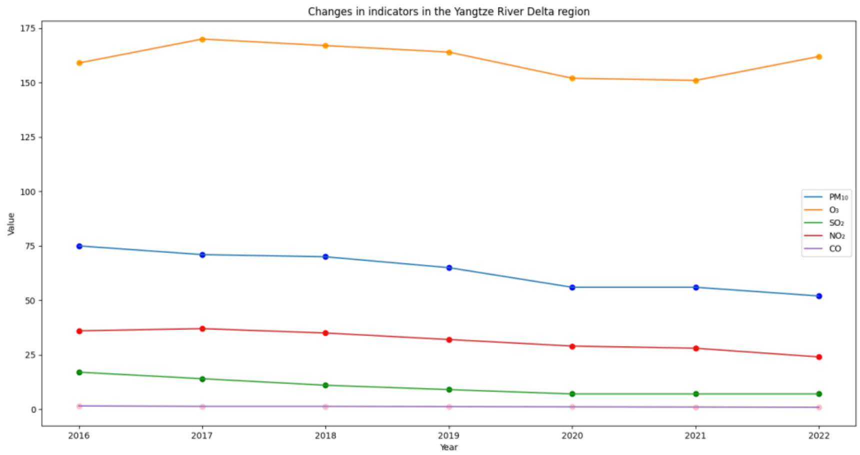
项目 / 年份	$PM_{2.5}$	较上年变化	PM_{10}	较上年变化	O_3	较上年变化	SO_2	较上年变化	NO_2	较上年变化	CO	较上年变化
2016	46	-13.20%	75	-9.60%	159	-2.50%	17	-19.00%	36	-2.7	1.5	0
2017	44	-4.30%	71	-5.30%	170	6.90%	14	-17.60%	37	2.80%	1.3	-13.30%
2018	44	-10.20%	70	-10.30%	167	0.60%	11	-26.70%	35	-5.40%	1.3	-7.10%
2019	41	-2.40%	65	-3.00%	164	7.20%	9	-10.00%	32	0	1.2	0
2020	35	-14.60%	56	-13.80%	152	-7.30%	7	-22.20%	29	-9.40%	1.1	-8.30%
2021	31	-11.40%	56	0	151	-0.70%	7	0	28	-3.40%	1.0	-9.10%
2022	31	0	52	-7.10%	162	7.30%	7	0	24	-14.30%	0.9	-10.00%

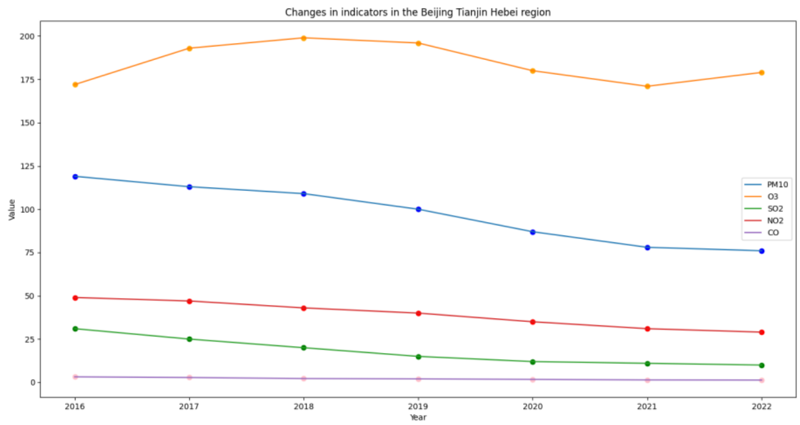
京津冀地区六项污染物浓度变化

项目 / 年份	$PM_{2.5}$	较上年变化	PM_{10}	较上年变化	O_3	较上年变化	SO_2	较上年变化	NO_2	较上年变化	CO	较上年变化
2016	71	-7.80%	119	-9.80%	172	6.20%	31	-18.40%	49	6.50%	3.2	-13.50%
2017	64	-9.90%	113	-4.25	193	12.20%	25	-19.40%	47	-4.10%	2.8	-12.50%
2018	60	-11.80%	109	-9.20%	199	0.50%	20	-31.00%	43	-8.50%	2.2	-24.10%
2019	57	-1.70%	100	-3.80%	196	7.70%	15	-16.70%	40	2.60%	2	0
2020	51	-10.50%	87	-13%	180	-8.20%	12	-20.00%	35	-12.50%	1.7	-15%
2021	43	-18.90%	78	-11.40%	171	-5.00%	11	-15.40%	31	-11.40%	1.4	-22.20%
2022	44	2.30%	76	-2.60%	179	4.70%	10	-9.10%	29	-6.50%	1.3	-7.10%

Secondly, in terms of data collection and analysis, we used tools such as Python and Excel for data processing and numerical calculations. Based on these data, we have also created charts and visualized results to present research conclusions and trend analysis more intuitively. It is worth noting that in the specific data analysis process, we have combined previous research results to make reasonable and operational constraints on data selection and analysis, to ensure the authenticity and reliability of the results.

Figure 3. Change curve of pollution indicators in two places.



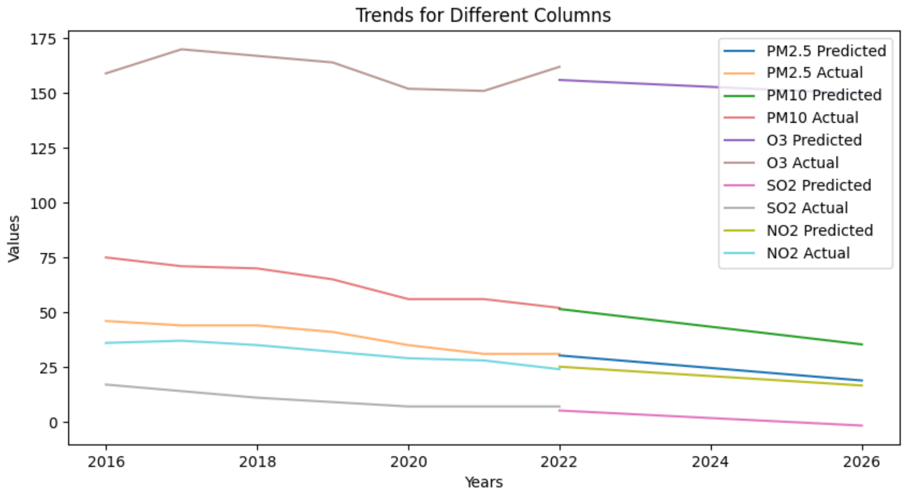


After the implementation of the new regulations on plastic restrictions, the attitude and intensity of urban environmental governance in various regions have gradually increased, and the results of urban environmental governance in 2020 have significantly improved compared to previous years. By analyzing the urban environmental data before and after 2020, we can conclude that the environmental governance effect in 2020 was significant, which to some extent reflects the effectiveness of the implementation of the new plastic restriction regulations.

2.2. Results and predictions

From the above data and charts, we can see that 2020 is the first year of the implementation of the new regulations, and significant results have been achieved in the control of urban pollution indicators in various regions during this year. At the same time, we conduct linear fitting predictions on the changes in pollution indicators in the coming years to further evaluate the impact of the new regulations on urban environmental governance.

Figure 4. Change trend curve and estimated curve.



3. Different participants and roles in the process

In the process of implementing the new regulations on plastic restriction, different participants in urban environmental improvement include government departments, plastic production enterprises, commercial retail groups, and ecological environmental protection organizations. They have taken different measures and actions to jointly promote the treatment of Plastic pollution and achieve the goal of improving the urban environment.

3.1. Government department

Government departments play an important role in the implementation of plastic restriction orders. In the process of implementing the new regulations, government departments actively promote and promote the new regulations, guide the response measures of all sectors of society, and strengthen supervision of industries related to plastic restrictions. For example, government departments have strengthened their inspection and law enforcement efforts against plastic factories and businesses, and strictly cracked down on and punished environmental pollution behaviors.

Figure 5. Prohibit plastic bags.

3.2. Plastic production enterprises

Plastic production enterprises are important participants in the implementation of plastic restriction orders. On the one hand, they have strengthened their control over plastic production and processing, and strengthened the production and sales of plastic products that meet the new regulations; On the other hand, large multinational companies such as Amazon and Unilever have begun to prioritize the use of alternative environmentally friendly materials, striving to promote sustainable industrial chains.

3.3. Commercial retail group

Commercial retail groups are another important participant in implementing plastic restriction orders. They effectively demonstrated how to use new packaging materials to replace traditional plastic bags and provide other sustainable products. For example, Wal Mart began to provide Reusable shopping bag, which has become an optional shopping method to promote the improvement of environmental awareness.

3.4. Ecological environment protection organization

Ecological and environmental protection organizations are also important participants in the implementation of plastic restriction orders. While supervising and promoting government departments, they actively carried out environmental education and publicity to improve public awareness and participation in Plastic pollution treatment. Some of these institutions will also carry out related scientific research projects to explore plastic treatment technologies and other alternative solutions for environmentally friendly materials.

3.5. Consumer

Consumers have a positive impact on the improvement of urban environment by choosing environmentally friendly products, reducing the consumption of plastic bags, and recycling waste plastics.After the implementation of the new regulations, consumers' purchasing behavior has changed. More and more people are choosing to use alternatives such as eco-friendly bags, eco-friendly cups, and eco-friendly tableware to reduce their dependence on plastic products. In addition, some consumers also regularly clean and classify garbage from their homes and offices, increasing the reuse rate of plastic resources.

Figure 6. Reuse of plastic resources.

4. Other aspects

4.1. Water environment

Prohibition or reduction actions are not only beneficial to the urban environment, but also very important for improving the water environment. According to a survey, PVC accounts for less than 5% of the entire plastic market share in China, but accounts for up to 50% of the proportion of abandoned plastic pipelines. Due to the long-term use of PVC pipes, they are prone to problems such as aging, fracture, and leakage, which can cause very serious pollution to the water quality of the water source. By limiting the production and use of plastic products, the quality of the water environment will be effectively improved. After the implementation of the plastic restriction order, some cities carried out water environment protection actions against Plastic pollution, and achieved remarkable results. For example, research on the relationship between large-scale landscape, ocean, inland waterway pollution, and summer plastic waste production shows that millions of tons of plastic waste enter the ocean every year, causing damage to the marine ecological environment. It is reported that 28 countries and regions have adopted legislation or policies to prohibit the production and use of disposable plastic products, forming an important trend in controlling global Plastic pollution.

4.2. Soil environment

Plastic waste is left in the soil for a long time and can slowly decompose due to factors such as light and heat, ultimately turning into harmful substances. According to the data of the Chinese Academy of Tropical Agricultural Sciences, 2.58 million tons of discarded agricultural film accumulated in rural areas of China, piled up in the field and occupied a large amount of farmland, which led to the early direct impact of water and soil loss in Hebei Province, known as the "wheat America". In addition, the production process of plastic products not only generates a large amount of carbon dioxide emissions, but also pollutes air, water sources, and soil. By reducing the use of plastic products, the impact of soil pollution can be effectively controlled.

5. Overview of Similar Research Achievements at Home and Abroad

In recent years, there have been many studies on the impact of plastic restriction orders on urban environmental governance in China. For example, Dou Wei and others (2009) discussed the plastic restriction order issued in 2008, and found that the implementation of plastic restriction order in China has made some phased progress in urban Plastic pollution control, but still faces difficulties in policy implementation.[1] The study also pointed out that in the implementation and promotion process of plastic restriction policies, government management departments, market participants, and consumers need to increase their initiative and enthusiasm for plastic restriction policies.

Some relevant studies abroad have shown that the implementation of plastic restriction orders is very effective and necessary for urban environmental pollution control. Policies such as restricting the use of disposable plastic products, encouraging the development of the environmental protection industry, and cultivating consumers' environmental awareness are effective measures.

Plastics have outgrown most man-made materials and have long been under environmental scrutiny. However, robust global information, particularly about their end-of-life fate, is lacking.As of 2015, approximately 6300 Mt of plastic waste had been generated, around 9% of which had been recycled, 12% was incinerated, and 79% was accumulated in landfills or the natural environment. If current production and waste management trends continue, roughly 12,000 Mt of plastic waste will be in landfills or in the natural environment by 2050.[2]

6. Discussion

The plastic restriction order is currently an important measure for urban environmental governance. This article mainly analyzes the urban atmosphere and explores the positive effects of plastic restriction measures on urban environmental governance from two aspects: water environment and soil environment governance. By reviewing relevant research and practice at home and abroad, it is found that the implementation of plastic restriction orders can reduce urban environmental pollution, protect water and soil environments, and is of great significance.However, we also need to pay attention to some issues. Firstly, although plastic restrictions can help improve urban environmental pollution, their scope of governance is still limited. Whether the comprehensive treatment and recycling of plastic waste can be truly achieved still needs further strengthening.

Secondly, strict regulations and high fines will be one of the considerations for some enterprises. It is also necessary to balance the relationship between environmental protection and economic development. For some small or high-risk enterprises, the government can take certain support or transitional measures to support their development of environmentally friendly products or further research on plastic substitutes, thereby promoting the development of the environmental protection industry.

In addition, the choices and behaviors of consumers are also crucial for the effectiveness of the implementation of plastic restriction orders. People should try their best to reduce the use of disposable plastic products, promote the Waste sorting and recycling lifestyle, and emphasize the concept of green consumption and sustainable development.

In summary, the implementation of plastic restriction orders is one of the correct directions to solve urban environmental pollution. With the joint efforts of government management departments, enterprises, and consumers, we believe that these measures will achieve more positive results.

References and Notes

1. Dou, Wei and Yong Wang. 2009. “Be Iimited to Mould the Countermeasure that Order ” Policy Analyses With the Effective Implementation[in Chinese](“Xiansuling” Zhengce Fenxi Yu Youxiao Shishi De Duice). *Science and Management*. *29*.02:43-46.
2. Geyer, R., Jambeck J. R., Law K. L. 2018. Production, use, and fate of all plastics ever made. *Science Advances* 4(7): e1700782.